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Correlation of the empirical polarity parameters of solvate ionic liquids (SILs) with molecular structure

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ABSTRACT

Empirical polarity parameters for 10 solvate ionic liquids (SILs) have been determined using Catalan's probes. The solvation environment surrounding the probe molecules in the SILs is dependent on the mutual interactions between the cation, anion and chelating ligand, leading to characteristic values of the polarity parameters. The acidity of the SILs is comparable to the acidity of polar protic solvents, which is attributed to the ability of the Li cation to interact with the probe in a manner similar to H-bond donor. The choice of anion influenced the measured acidity of SILs by determining the extent of cation–probe interactions possible. The basicity showed a strong correlation with the nature of the anion but was also influenced by the extent of cation-anion interaction and choice of ligand. Temperature dependence of polarity parameters in SILs is relatively small, but shows some interesting trends.

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1. Introduction

Solvate ionic liquids (SILs)[1], which are equimolar mixtures of lithium salts with solvents, are emerging as a promising subclass of ionic liquids since they were first synthesized by Pappenfus et. al. [2]. SILs display all the desirable traits of ionic liquids such as low vapour pressure, thermal stability, wide electrochemical window and low flammability. Research in the field of rechargeable batteries has been gaining momentum with the increasing need for alternative sources of energy [3]. The need for more efficient electrolytes in rechargeable batteries is the primary driving force for the increasing popularity of SILs [4]. Recently, SILs have been utilized as reaction media for organic reactions with encouraging results [5]. The efficiency of SILs for *in vivo* delivery of therapeutics and as sizing agents for carbon fibres have also been explored [6,7].

The characteristic feature for a typical equimolar mixture of (lithium salt + solvent) to be classified as a SIL is the existence of long-lived solvated Li cation [8]. The solvents which are good chelating agents typically employed for the purpose are glycols and glymes. Weakly coordinating anions also facilitate stronger interaction between the cation and the complexing solvent, resulting in properties entirely different from those expected for a "concentrated solution" of Li salt in organic media [9]. The extent of solvation and its effect on the properties of the mixture have been extensively studied and discussed [10]. The role of the cation and anion in stabilising the solvated cation conformers of SILs has been investigated through ab initio calculations [11]. The influence of the local environment around the cationic and anionic species on the transport properties of SILs has been investigated through molecular modelling [12].

Such molecular level perspective is critical for understanding the physicochemical properties of SILs such as viscosity, ionicity and conductivity [13]. In addition, the nature of intermolecular interactions in SILs with dissolved solutes also needs to be examined thoroughly for ensuring optimum utilization of SILs as reaction media. Polarity of SILs, for example, can be the key factor in determining the outcome of a chemical process carried out in these systems. In one of the earliest reports on the polarity of ionic liquids, Eyckens and coworkers determined the E_T^N and Kamlet–Taft polarity parameters for [Li (G3)][NTf_2] and [Li (G4)][NTf_2] and compared them to the pure solvent components G3 and G4, respectively [14]. This was followed by a more extensive study in which Kamlet–Taft polarity parameters for a set of 20 Li-based SILs was determined [15]. It was observed that the values of the polarity parameters in SILs was greatly influenced by the stability of the solvated cation, with the "poor" SILs having β and π* values similar to the parent glyme or glycol. It was also reported that the use of Reichardt's dye was not feasible for all SIL samples studied, thus necessitating the use of Burgess' dye. Kamlet–Taft parameters for Li and glyme mixtures with different compositions (other than the equimolar composition) have been reported along with NMR spectra and thermochemical studies [16]. A major limitation of the previous approaches for determining the polarity of solvents is the use of molecular probes did not ensure a generalised / universal scale of solute – solvent interactions. The interactions of the probe with the solvent need not differentiate the specific and nonspecific solvent interactions, giving rise to polarity scales which were subject to the choice of probe molecules. The

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multiparameter approach proposed by Catalan and coworkers overcomes these limitations and has the advantage that it leads to absolute polarity values derived by the use of homomorphic probes rather than the relative values resulting from use of conventional polarity probes [17–21]. The values of solvent acidity, basicity, polarizability and dipolarity resulting from Catalan's approach are well-defined, unlike previous attempts which are only capable of determining the 'average' solvatochromic parameter, invariably influenced by the contribution of other parameters. Catalan's method is also instrumental in separating the relative contribution of solvent dipolarity and polarizability. Till date, Catalan's probes have been successfully employed to measure the polarity of organic solvents[22], ionic liquids [23] and deep eutectic solvents [24]. There has been no report discussing the polarity of SILs in terms of the generalised polarity scale employing Catalan's approach.

Herein we determine the polarity parameters of various solvated ionic liquids using Catalan's generalised scale of polarity as compared to previously reported Kamlet–Taft parameters (Scheme 1). All the SILs studied are derived from lithium salts with anions of different coordination ability – trifluroacetate ($[TFA]^-$), trifluromethanesulfonate ($[OTf]^-$) and bis (trifluromethylsulfonyl)imide ($[NTf_2]^-$). The range of chelating solvents employed in the present study includes triglyme (G3) and tetraglyme (G4) as well as corresponding alcohols, triethylene glycol (E3) and tetraethylene glycol (E4). The stability of the $[Li(glyme)]^+$ complex varies in the order – $[TFA]^- < [OTf]^- < [NTf_2]^-$ and that is the expected order of stability as "solvate" ionic liquids. The choice of the anion and chelating components was made with a view to encompass the range of SILs –"good" SILs to "poor" SILs. The effect of temperature on the polarity parameters has also been discussed for four representative solvate ionic liquids. The results highlight the unique nature of this class of "solvents" in terms of the range of polarity observed which will have important implications for future applications of SILs.

2. Materials and methods

2.1. Materials

The purity of the triglyme, tetraglyme, triehtylene glycol and tetraethylene glycol used in the present work was greater than 99%. Commercially available lithium salts with >99% purity were used without further purification. Anthracene, 5-nitroindoline (NI) and 1-methyl-5-nitroindoline (MNI)[19] used for spectroscopic determination of polarity were of highest purity available. All the molecular solvents used for standardising the polarity measurements were of spectroscopic grade.

2.2. General procedure for synthesis of ionic liquids

A series of glyme based ionic liquid were prepared and characterized as previously reported [10c]. Equimolar amounts of glyme (or glycol) and Li salt were mixed under nitrogen atmosphere and stirred for 24 h at 60 °C until colorless and homogeneous liquids were obtained. The purity of ionic liquid samples was confirmed using H [1] NMR and CHN analyses (details provided in Supporting Information.) The SILs were stored in vacuum desiccators to maintain moisture free conditions and were thoroughly dried before every use.

2.3. Synthesis of polarity probes

The molecular probes i.e. 2-fluro-7-nitrofluorene (FNF)[25], 2-*N*,*N*-dimethylamino-7-nitrofluorene (DMANF) [17,18], 3,6- diethyltetrazine (DETZ) [26] were synthesized as per reported procedures and characterized by NMR (see Supporting Information).

2.4. Measurement of polarity

Solvatochromic parameters were determined for 5 molecular solvents and compared with literature values (% deviation from reported values < 2%) in order to standardise the spectroscopic procedure. Samples of SILs were thoroughly dried in vacuum at >70 °C for 2 h before every measurement to ensure complete drying. Freshly prepared stock solution of the dye in dichloromethane was added dropwise to the solvate ionic liquids and solvent was evaporated under vacuum. The solvated ionic liquid sample was then transferred to the cuvette under nitrogen atmosphere and sealed with septum. The λ_{max} (wavelength accuracy and reproducibility $= \pm 0.1$ nm) was measured at different temperature using UV–Visible spectrophotometer equipped with a Peltier temperature controller having accuracy of ± 0.1 K. Multipeak fitting was used to determine the λ_{max} with greater accuracy. All spectroscopic measurements were performed at least in triplicate starting from sample preparation, and the results were averaged (standard deviation < 2%). Temperature dependent spectroscopic measurements were carried out at by varying the temperature from 298 to 318 K (interval of 5 K).

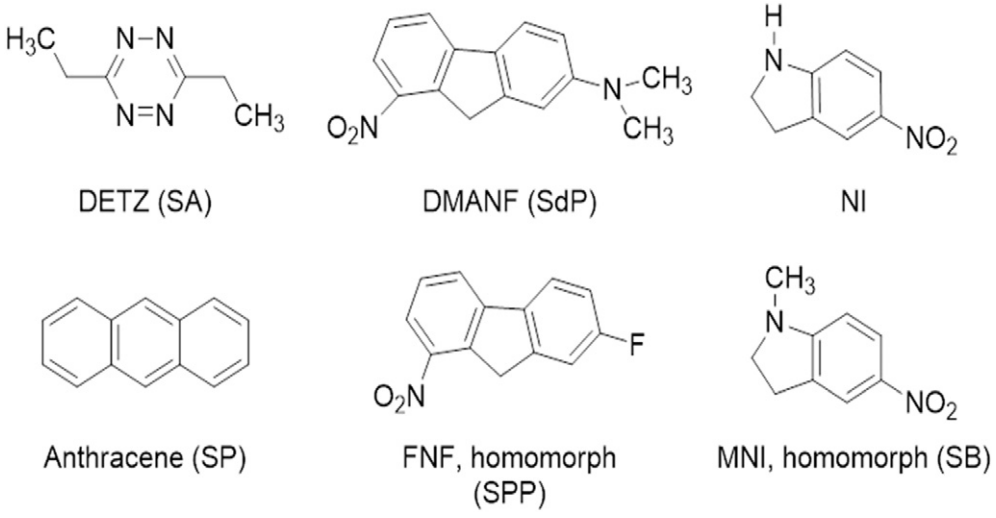
3. Results and discussion

The four independent solvatochromic parameters (SP, SdP, SA and SB) were initially determined for ten SILs at 298 K using Catalan's probes as summarized in Scheme 1 and the resulting values are listed in Table 1. All the dyes were stable in the SILs and no deterioration / decrease in intensity was observed during the time course of the measurements. Analogous measurements were carried out for the four chelating solvents (E3, E4, G3 and G4) in order to facilitate comparison and to analyse the contribution from the chelating agent. Previously reported data for molecular solvents, ionic liquids and deep eutectic solvents has also been listed to facilitate a comparative discussion in the following subsections.

3.1. SP parameter

The polarizability of the SILs (SP parameter) was determined by monitoring the shift in the absorption bands of anthracene, since its spectral behaviour is primarily influenced by the solvent polarizability. The polarizability data can be validated by comparison with the corresponding values estimated using the Lorentz—Lorenz equation. The equation estimates the polarizability of the solvent as a function of the refractive index. In order to apply the equation, temperature dependent refractive index values are required, which are available only for $[Li(G4)][NTf_2]$ till date [27]. Experimental SP values show a strong linear correlation with the Lorentz–Lorenz polarizability function (Fig. 1, $n = 5$, $r = 0.988$ and $SD = 0.0006$), validating the estimated polarizability scale at least partially. Lack of temperature dependent refractive index data prevents an analogous treatment for the rest of the SILs analysed in the present work.

In general, it is observed that the SP values for SILs composed of triglyme or triethylene glycol ligands are lower than those observed for SILs containing tetraglyme



Scheme 1. Probe molecules used for polarity determination.

Table 1
Empirical polarity parameters using Catalan's probes at 298 K.

Solvent	SP	SdP	SA	SB	SPP
[Li (G3)][NTf_2]	0.677	0.921	0.600	0.106	0.937
[Li (G4)][NTf_2]	0.694	0.920	0.561	0.270	0.984
[Li (G3)][OTf]	0.713	0.802	0.190	0.133	1.227
[Li (G4)][OTf]	0.720	0.906	0.337	0.480	1.033
[Li (E3)][NTf_2]	0.671	1.021	1.19	0.112	0.644
[Li (E4)][NTf_2]	0.725	0.840	0.704	0.150	0.928
[Li (E3)][OTf]	0.730	0.890	0.782	0.230	0.905
[Li (E4)][OTf]	0.750	0.780	0.776	0.394	0.815
[Li (E3)][TFA]	0.690	0.800	0.584	0.133	0.987
[Li (E4)][TFA]	0.751	0.670	0.712	0.190	0.894
Triethylene glycol (E3)	0.768	0.846	0.378	0.530	0.991
Tetraethylene glycol (E4)	0.762	0.869	0.332	0.573	1.017
Triglyme (G3)	0.723	0.667	0.324	0.568	0.874
Tetraglyme (G4)	0.732	0.639	0.320	0.622	0.859
[BMIM][BF_4][a]	0.783	0.966	0.528	0.381	0.917
[BMIM][PF_6][a]	0.728	0.998	0.529	0.21	0.986
Reline (ChCl + urea, 1:2)[b]	0.887	0.721	0.677	0.359	0.963
Ethaline (ChCl + ethylene glycol, 1:2)[b]	0.816	0.975	0.546	0.379	0.966
Glyceline (ChCl + glycerol, 1:2) [b]	0.851	0.878	0.583	0.436	0.975
Pentane[c]	0.593	0	0	0.073	0.507
2-methylbutane[c]	0.581	0	0	0.053	0.479
perfluorohexane[c]	0.339	0	0	0.057	0.214
1-bromonaphthalene[c]	0.964	0.669	0	0.202	–
2-hexanol[c]	0.683	0.601	0.14	0.966	0.487
2-octanol[c]	0.696	0.496	0.088	0.963	0.786
1, 2, 3-propanetriol[c]	0.828	0.921	0.653	0.309	0.948
1, 2-ethanediol[c]	0.777	0.91	0.717	0.534	0.932
aniline[c]	0.924	0.956	0.132	0.264	–
water[c]	0.681	0.997	1.062	0.025	0.962
Formamide[c]	0.814	1.006	0.549	0.414	0.833
Acetic acid[c]	0.651	0.676	0.689	0.39	0.781

[a] Data from reference 23.
[b] ChCl = choline chloride, data from reference 24.
[c] Data from reference 22.

or tetraethylene glycol. The SP value of [Li (E3)][NTf_2] is the lowest (0.671) in glycol based ionic liquids while the SP value of [Li (G3)][NTf_2] is lowest among the glyme containing ionic liquids. This is not necessarily the case for the pure glyme or glycol components. For example, the SP value of E3 is higher than that observed for E4 as a solvent.

Within the glyme based SILs, the polarizability is sensitive to the identity of the anion and follows the order: NTf_2 < TFA, i.e. SP value is lower for a typical "good" SIL and higher for "poor" SIL or "concentrated solutions". For glycol-containing SILs, the trend in SP is not consistent. In SILs with the [Li (E4)]$^+$ cation i.e. tetraethylene glycol as the chelating component, solvent polarizability varies with the anion in the order $[NTf_2]^-$ < $[TFA]^-$ ≈ $[OTf]^-$. If the cationic species is composed of the triethylene glycol component i.e. [Li (E3)]$^+$, the order changes to $[NTf_2]^-$ < $[TFA]^-$ < $[OTf]^-$. This is an indication that the polarizability of the medium (as experienced by anthracene) is a complex outcome of interactions with both the cation and anion of the solvate ILs.

The values of the SP parameter for all the 10 SILs included in this study are lower than the corresponding 'chelating' component (pure glycols and glymes). With respect to the molecular solvents, it is observed that the SP values are 'intermediate' between the nonpolar and polar solvents. The SP values for the imidazolium ionic liquids, are much higher – probably due to the aromatic nature of the imidazolium ring. The SP values for deep eutectic solvents are also higher than those observed for SILs in the present report. This may indicate a stronger interaction between the components (cation, chelating compound and anion) in solvate ILs as compared to deep eutectic solvents, such that the stabilising influence of either on the electronic transition of the probe molecule is reduced.

It must be noted that the SP need not correspond to π* parameter from the Kamlet Taft scale, since the latter includes contribution from dipolarity as well as polarizability which cannot be separated using the conventional approach. The π* values can expressed as a linear combination of the independent solvatochromic parameters for molecular solvents and ionic liquids. This possibility will be explored for solvate ILs in the following sections.

3.2. SdP parameter

Solvent dipolarity (SdP parameter) is another solute–solvent interaction term (apart from polarizability), which is non-specific. It is measured by solvatochromic comparison of two probe molecules – 2-*N*,*N*-dimethylamino-7-nitrofluorene (DMANF in Scheme 1) versus anthracene. One of the major accomplishments of the Catalan's approach of empirical polarity determination is the ability to delineate contributions from solvent polarizability and dipolarity to the non-specific solute–solvent interactions. Validation of the empirically obtained dipolarity values can be done through correlation with the Kirkwood dipolarity values. However, paucity of experimental data related to the static permittivity of SILs excludes this option.

The solvent dipolarity (SdP) values consistently show highest values for the $[NTf_2]^-$ anion for all the classes of chelating ligands and lowest values for the $[TFA]^-$ anions depends on the identity of the chelating ligand. The solvent dipolarity in SILs thus is predominantly dependent on the coordinating ability of the anions. Lower values of SdP may result from the combination of a poor chelating agent and a strongly coordinating anion, which lead to a "concentrated solution" of Li-salt and glycol / glyme rather than "solvated Li species". SILs in which the cation-anion pairs are strongly coordinated are observed to be less capable of stabilising the excited state dipole of DMANF. In such cases, the probe molecule may also be subject to stabilisation by the "free" chelating species (glyme or glycol) but the extent of stabilisation may be less. In contrast, the weakly coordinated cationic and anionic species in "good" SILs may act in tandem to stabilise the probe. In general, SdP values for glyme-based SILs are higher than those observed for the pure glymes, Among the glycol-based SILs, E3 has lower SdP as compared to the corresponding SILs while E4 has higher dipolarity as compared to the SILs composed of E4.

The range of SdP observed in SILs is comparable to the values observed for moderately polar to very polar molecular solvent, with a few notable exceptions. The solvent dipolarity values for ionic liquids reported in literature are high and are comparable to the values observed for glyme-based SILs, whereas glycol-based SILs show much lower dipolarity (with the exception of [Li (E3)][NTf_2]). The SdP values for DESs show much comparable variation in values, although the literature is limited on this class of solvents.

3.3. SPP parameter

The polarizability and dipolarity parameters should ideally yield an overall polarity descriptor for non-specific solute–solvent interactions. In Catalan's approach towards measurement of empirical polarity, this factor is designated as the 'solvent polarity / polarizability' parameter (SPP), which is empirically determined by the solvatochromic comparison of DMANF and FNF probe molecules. For conventional solvents and ionic liquids, the empirical solvent polarity and polarizability scale (SPP_{exp}) shows a linear correlation with the theoretical SPP values (SPP_{calc}) obtained by a linear correlation of the SP and SdP parameters given as $SPP_{calc} = (0.49 \pm 0.03)\,SP + (0.38 \pm 0.01)\,SdP + (0.21 \pm 0.02)$.

No correlation was observed for the SPP parameters derived for the 10 at 298 K. In the case of solvate ILs, temperature-dependent polarity data for three out of the four SILs yielded a linear correlation between the SPP_{exp} and SPP_{calc} but the slope was not necessarily close to unity. For example, the correlation between the experimental and theoretical SPP values in [Li (G3)][NTf_2] is consistent with the trend in molecular solvents (Table 2). However, the correlation was poor for the other SILs examined – namely [Li (G3)][OTf], [Li (G4)][OTf] and [Li (G4)][NTf_2]. This indicates that the relative contribution of the solvent dipolarity and polarizability may vary in the case of solvate ILs. One of the reasons for this deviation is the fact that the "structure" of SILs is dynamic in terms of the interactions between the components.

The SPP values for the glyme-based ionic liquids are higher than those for pure glymes while the SPP values of glycol-based SILs are slightly lower than the component glycols. The SPP values for ionic liquids and DES are comparable to those observed for SILs.

The major advantage of computing the SPP parameter is that it can be compared to the Kamlet-Taft π* parameter to derive qualitative information about the nature of solvent interactions. Although it is difficult to derive an overall correspondence between the SPP and

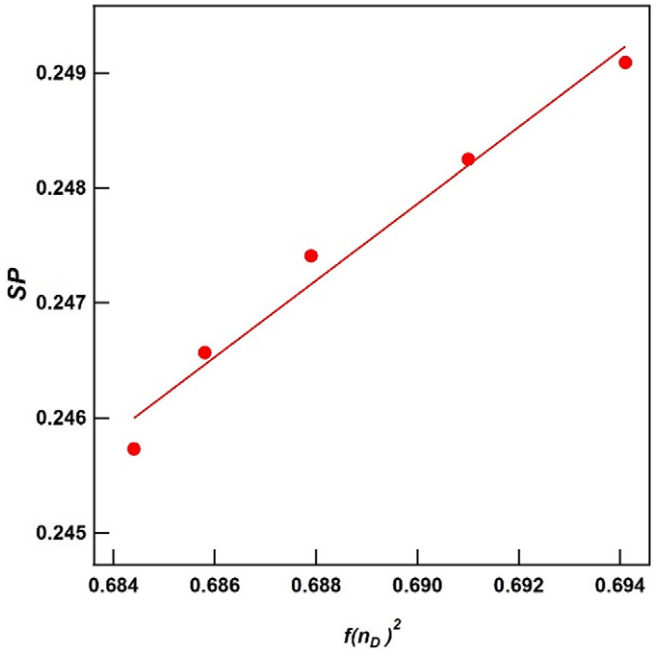


Fig. 1. Correlation between temperature-dependent SP scale and Lorentz−Lorenz function for [Li (G4)][NTf_2].

Table 2
Correlation parameters for $SPP_{exp} = m.SPP_{calc} + c$ between 298 K and 318 K.

SIL	*m*	*c*	*n*	*r*	SD
[Li (G3)][NTf_2]	1.03 ± 0.06	0.01 ± 0.05	5	0.89	0.009
[Li (G4)][NTf_2]	− 0.5 ± 0.2	1.4 ± 0.2	5	0.77	0.05
[Li (E3)][NTf_2]	2.46 ± 1.04	− 0.9 ± 0.9	5	0.80	2.05
[Li (E4)][NTf_2]	2.22 ± 1.04	− 0.9 ± 0.9	5	0.56	2.14

π* values, some general observations can be made. For example, the variation in SPP parameter for SILs with change in the chelating component is comparable for all anions ($[NTf_2]^-$ and $[OTf]^-$). However, the π* values for the SILs containing the $[OTf]^-$ anion are almost invariant and do not vary significantly when the chelating component is varied. This indicates that the magnitude of the π* component may be predominantly influenced by the identity of the anion while the SPP parameter is not biased towards any single structural factor. Catalan's SPP parameter may be a more appropriate indicator of non-specific solvent polarity.

3.4. SA parameter

Solvent acidity is a measure of the Bronsted acidity of the solvent and is determined by modifying the spectroscopic behaviour of DETZ (Scheme 1) to eliminate the non-specific interactions using the SPP parameter (See Supporting Information). Acidity of SILs is higher than that observed for pure glymes and glycols, except [Li (G3)][OTf] and [Li (G4)][OTf]. On an average, glycol based SILs have higher acidity than the glyme based SILs. The glyme based ionic liquids are particularly sensitive to the nature of the anion with G3 being more sensitive. For example, when the anion is changed from [OTf] to [NTf_2], the SA values increase by 300% for $[Li (G3)]^+$. The higher acidity value is then a result of the stabilisation due to "Li-bonding" similar in nature to H-bonding, as discussed in literature [14,15]. Substitution of a strongly coordinating anion with a weakly coordinating anion enables greater interaction between the chelated L-cation and the hydrogen bond acceptor on the probe molecule. According to our results, the interaction between the probe and the terminal OH groups in glycol may also become pertinent in SILs with glycol constituent.

On an average, the SA values for SILs fall in the range associated with polar protic solvents such as 1,2-ethanediol, 1, 2, 3-propranetriol and water. The observed acidity values are comparable to those seen for ionic liquids or deep eutectic solvents. The similarity in magnitude do not imply uniformity in the nature of molecular interactions. For example, acidity of the imidazolium –based ionic liquids primarily arises due to the interactions of C2–H (or the acidic hydrogen on the imidazolium ring) In SILs, however the observed acidity is a result of Li-interactions with the probe molecules.

For glyme based SILs, acidity of NTf_2 based SILs is greater than TFA containing SILs, in general. This is also observed for the α values reported by Dolan et.al. [15]. Weak cation-anion coordination may result in better access to the solvated Li^+ ion for the probe. It must be noted that the previous report on the α values was based on the solvatochromic behaviour of Burgess' dye due to the low solubility of the Reichardt's dye in SILs. The present study does not face any such limitations.

3.5. SB parameter

The basicity of solvents is represented through the SB parameter using the MNI probe and its homomorphy NI. The resulting values for the SILs and other relevant solvents are listed in Table 1. The basicity of SILs as reflected in the SB values, is lower than that observed for the pure components and even DESs. The solvent basicity is also influenced by the nature of the anion and follows the order NTf_2 < TFA < OTf. This is remarkable since the coordinating ability of the anions is expected to be NTf_2 < OTf < TFA, which denotes the transition of the system from "good" SILs to "poor" SILs (or concentrated solutions). The fact that the SB values are neither directly or inversely proportional to the expected trend indicates that solvent basicity is not solely dependent on the anionic properties and more than one contributing factors may be influencing the outcome. For example, strongly basic anions such as $[TFA]^-$ anions are also strongly coordinating. The strong cation–anion coordination and the resulting significant ion-pair formation with Li ions, can be detrimental to the basicity of $[TFA]^-$ anion [28]. Thus moderately basic OTf showed the highest basicity. The oxygen atoms on the glycol or glyme can also contribute to the observed Bronsted basicity of SILs. The SB values of SILs are much lower than those observed for the pure glyme or glycol. This may be due to fact that the oxygen atoms which act as the primary hydrogen bond acceptor are now coordinated to the Li cation and cannot contribute towards stabilising the hydrogen bond donor site in the probe molecule. For a given anion, the basicity apparently increases in the order G3 < E3 < E4 < G4, which may be reflection of the stability of the Li–solvate stability. Greater the extent of chelation for the cation, lower is the extent of cation–anion pair formation, thus leading to increased basicity. But, it is difficult to reach a conclusion in absence of data for all the SILs involved. The Kamlet-Taft β values reported in literature are much higher the SB parameter.

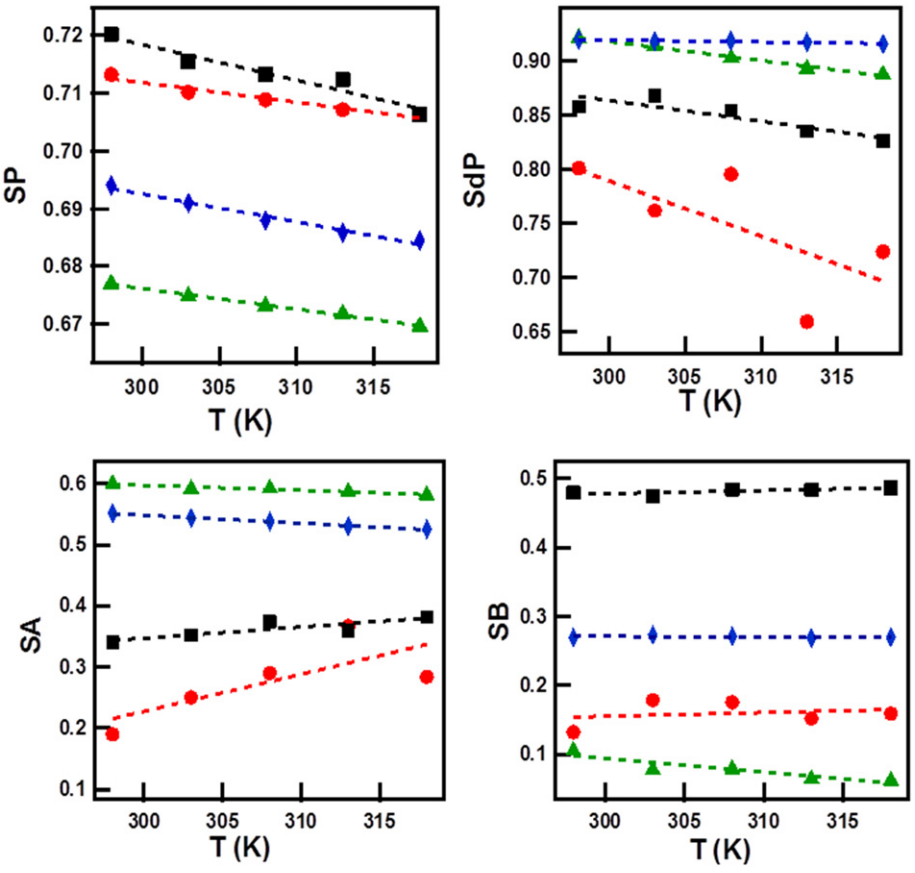


Fig. 2. Temperature-dependent response of SP, SdP, SA, and SB parameters for [Li (G3)][NTf_2] (▲), [Li (G4)][NTf_2] (♦), [Li (G3)][OTf] (●) and [Li (G4)][OTf] (■).

3.6. Temperature dependence of polarity

The variation of the SA, SB, SP and SdP parameters with temperature were recorded for four SILs – namely [Li (G3)][NTf_2], [Li (G4)][NTf_2], [Li (G3)][OTf] and [Li (G4)][OTf] and the results are summarized in Fig. 2. It can be observed that, barring a few exceptions, the polarity of SILs shows negligible temperature dependence. In the cases where temperature dependent polarity changes are observed, it is mostly in the form of an inversely proportional relation between the magnitude of polarity parameter and temperature. This indicates weaker interactions between the probe and the solvent medium at higher temperatures. The linear temperature dependence of SILs is consistent with that observed for molecular solvent, most of which show a negligible to marginally negative temperature dependence. Another important observation is that the extent of temperature-dependent variation in polarity of the SILS depends on the identity of both – the solvate cation and the anion. No clear cation or anion dominance is observed in determining the fate of the polarity parameters as a function of temperature.

4. Conclusions

The polarity of SILs is observed to be an interesting outcome of the interaction between the cation, chelating species and anion. The weakly coordinating anions are known to promote cation-glyme / cation/glycol interactions and have characteristically different polarity behaviour. However, the extent of cation-anion coordination is not the only dominant factor and there can be other molecular interactions which contribute to the observed polarity values, depending on the nature of the probes employed. It should be difficult to categorise the solute-solvent interactions in SILs based on the classification of SILs or the properties of the components that constitute the SILs. The temperature dependence of polarity in SILs is consistent with that observed for conventional solvents and does not deviate from linearity in the range studied. The choice of SILs for a particular application based on the polarity parameters determined at room temperature will be robust for a reasonable range of working temperature.

Declaration of competing interest

The authors have no conflicts of interest to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2019.111882.

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